# Combinational Logic – Part 1

K. Sridharan

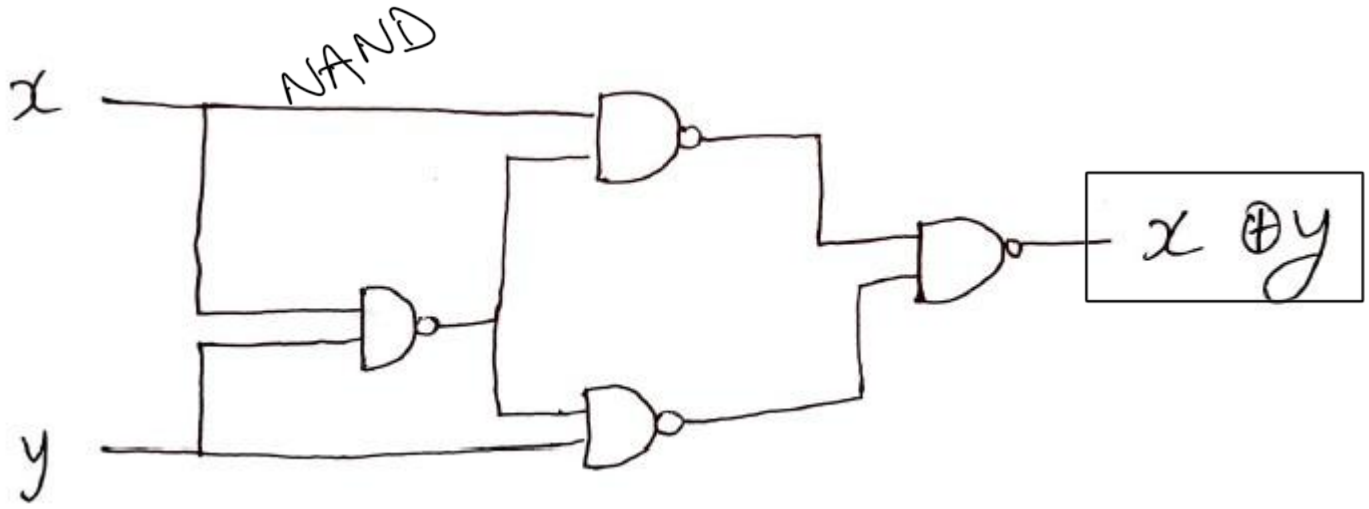
# Review

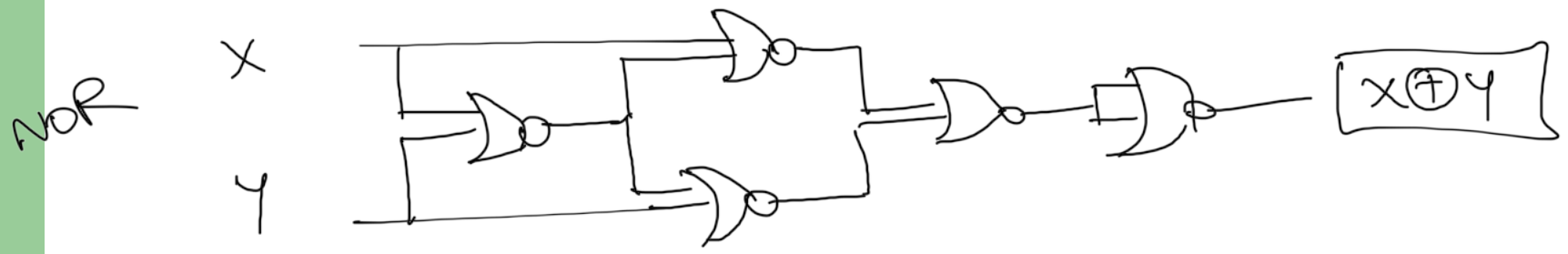
- **We have discussed logic minimization.**
- **We will now begin discussion of various combinational logic circuits (the adders we mentioned earlier are also combinational logic elements).**

# What is a combinational circuit ?

- **A combinational circuit consists of logic gates whose outputs at any time are determined from only the present combination of inputs.**
- **A combinational circuit performs an operation that can be specified by Boolean logic functions.**

# Example of a combinational circuit



# Various combinational logic elements

- **A variety of combinational circuits for different purposes can be given.**
- **These include adders, subtractors, encoders, decoders, comparators and so on.**
- **We will begin with adders.**

# Two forms of 1-bit adders

- **We can have a binary adder that adds two bits at a time or a binary adder that adds three bits at a time.**
- **The former is called as a half adder (or half-adder) while the latter is called as a full adder (or full-adder).**

# A half adder

Takes two binary inputs (a, b) and produces two binary outputs (s, c).

| a | b | s | c |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

# More on the half adder

- **The inputs to the half adder are a and b while the outputs are s and c.**
- **We use the symbols s and c to denote sum and carry respectively.**
- **Note that the sum s is actually the XOR of the inputs while the carry c is the AND of the inputs.**

# Full adder

| x | y | z | s | c |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

# On the full adder truth table

- **Here, we have three inputs (a, b, c) and two outputs (s, c).**
- **The sum (s) is given by a XOR b XOR c (Verify !)**
- **The carry (c) is given by ab + bc + ca (Verify !)**

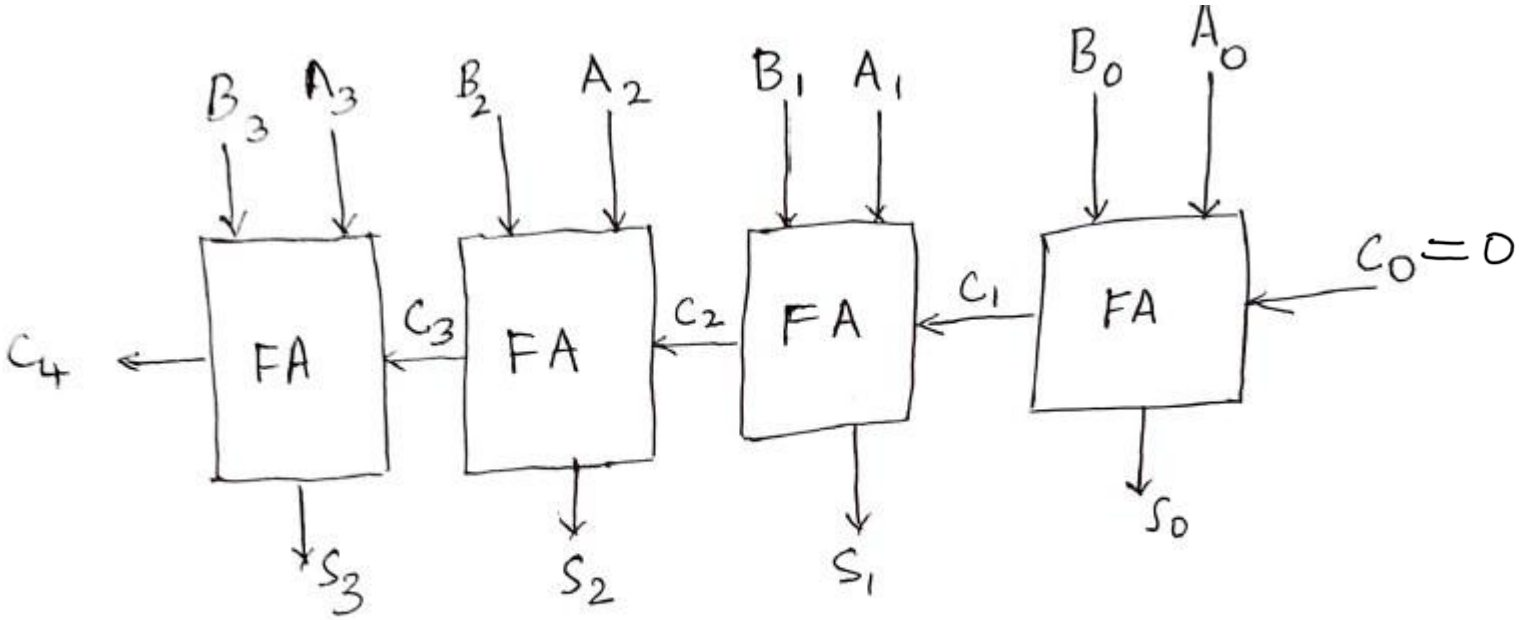
$\Sigma m(1,2,4,7)$

$\Sigma m(3,5,6,7)$

# How to do multi-bit addition ?

- **The full adder is valuable when we want to add two multibit numbers.**
- **Different ways to add two numbers (each of which has many bits) are possible.**
- **Some solutions are fast, some require less hardware and so on.**

# Low hardware solution (called ripple carry addition) for 4-bit numbers




FA: Full adder

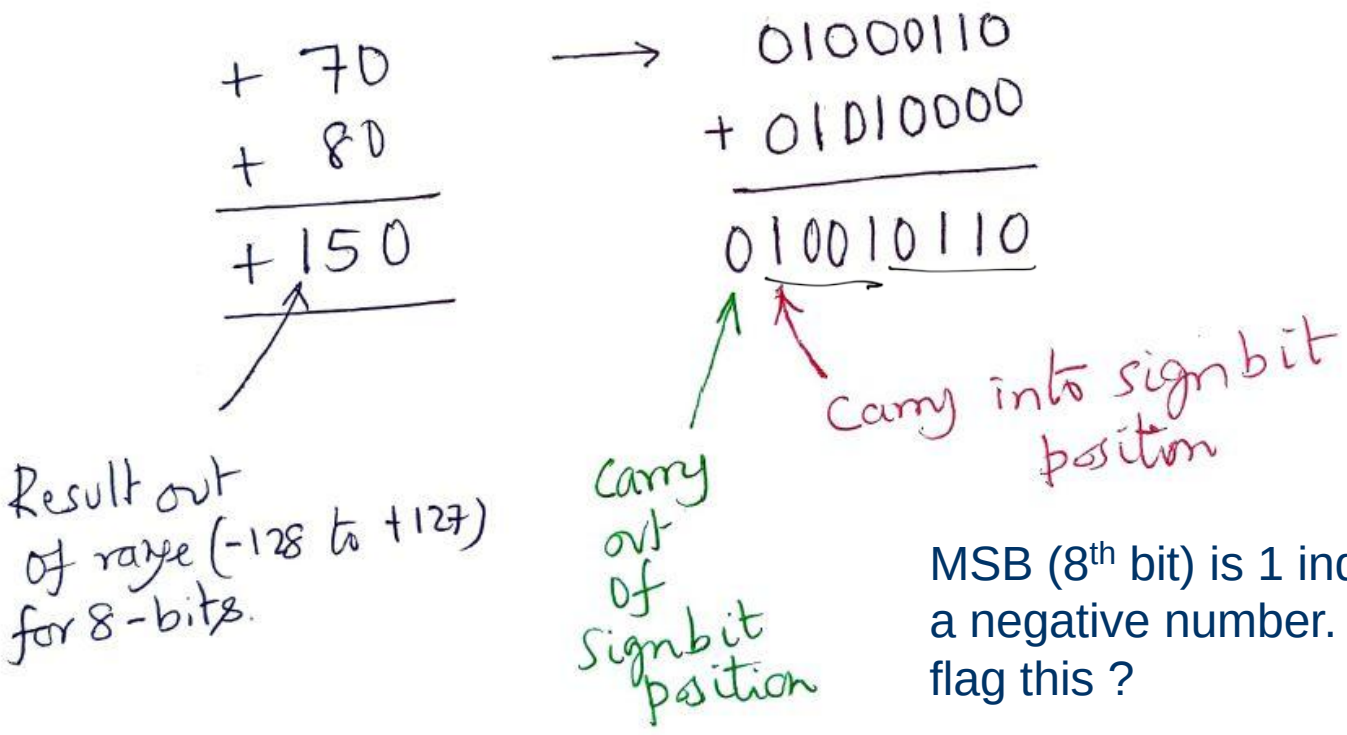
# Can we extend to subtraction ?

- **Yes. In fact, subtraction of unsigned binary numbers, A and B can be done via 2's complement representation of the number to be subtracted.**
- **That is, to find A – B, just take 2's complement of B and add to A. The 2's complement of B can be obtained using inverters (to realize 1's complement) and addition of 1 to the least significant bit.**

# Can we have a single circuit for addition and subtraction ?

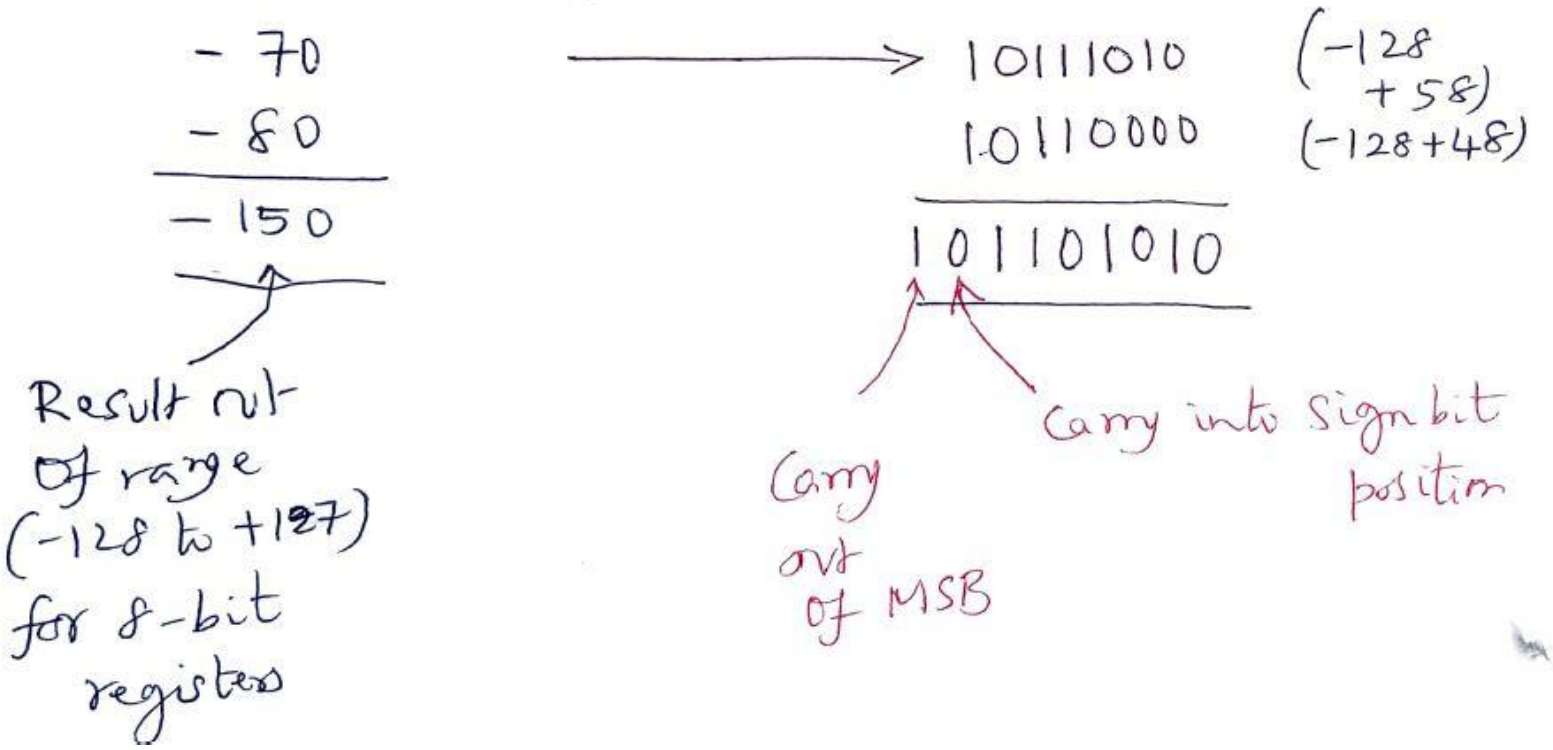
- **Yes. Before we see the circuit, we need to understand the notion of Overflow.**
- **For unsigned number addition, an Overflow is detected from the end carry out of the most significant bit position.**
- **For signed numbers in 2's complement form, note that there is a "smaller" range of valid numbers given a certain size for the registers that hold the input numbers. If we assume 8-bit registers, then addition of +70 to +80 will cause an overflow. Similarly, addition of -70 to -80 will also cause an overflow. We will see both of these next.**

# The input numbers are positive but the result exceeds the capacity of an 8-bit register




MSB (8th bit) is 1 indicating a negative number. How to flag this ?

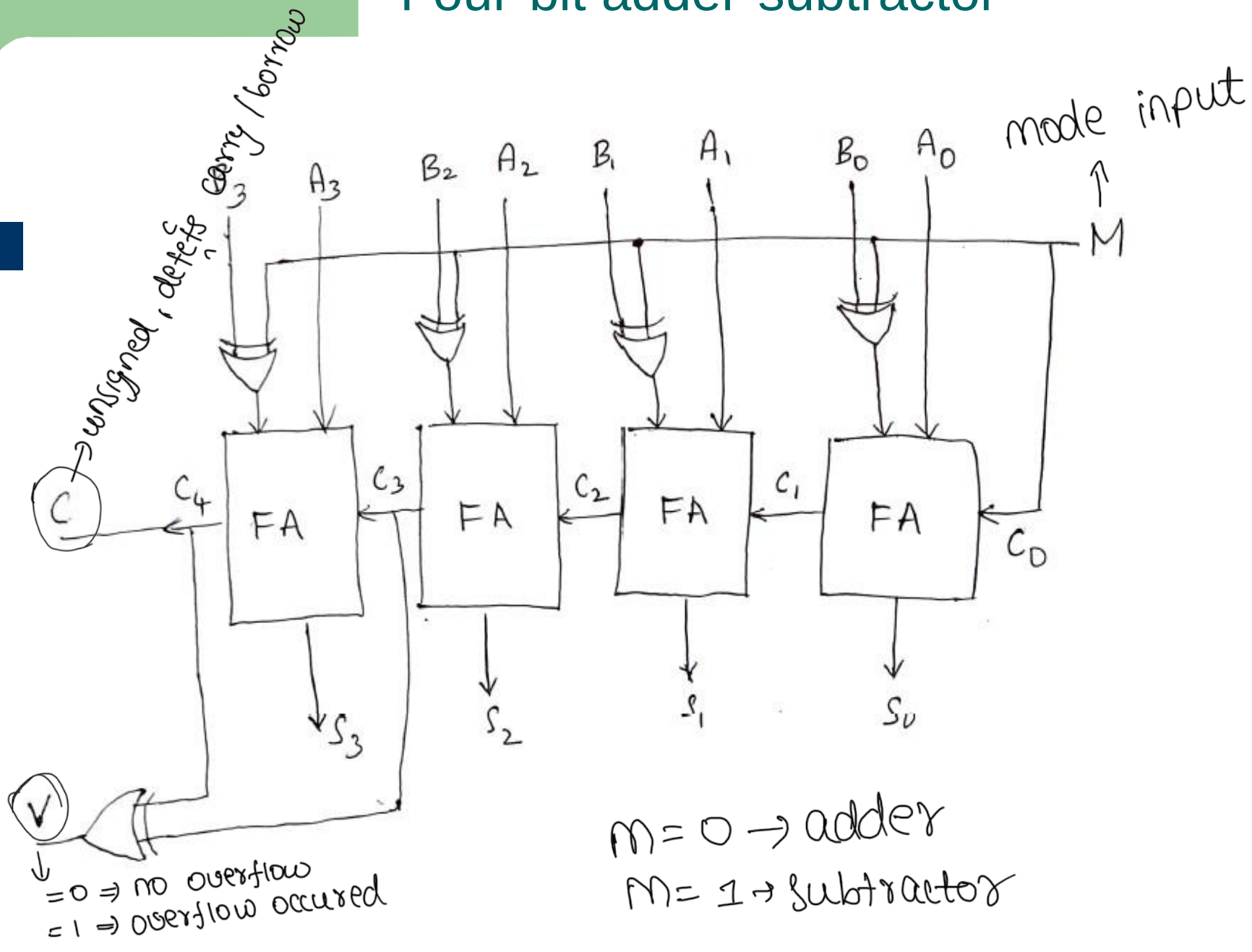
# Another scenario: The input numbers are negative but the result again is out of range



# Overflow

- **In the signed case, whenever the result exceeds the range, we can observe that the answer (restricted to the output register capacity) is incorrect. This needs to be flagged somehow.**
- **This is done by comparing the carry into the sign bit position with the carry out of the sign bit position. If the two carries are not equal, we say an overflow has occurred. We now look at the four bit adder-subtractor circuit that also has an overflow output (denoted by V).**

# Four-bit adder-subtractor



# Explanation

- **Addition and subtraction operations are combined into a single circuit by incorporating an XOR gate with each full adder.**
- **The 4-bit adder-subtractor circuit has a mode input denoted by M. When M=0, the circuit is an adder (does A + B) while when M = 1, the circuit performs A – B.**
- **Note that for the full adders, the input carry is 0. For subtraction, we have the B input (bits) complemented and a 1 is added through the input carry so A – B is accomplished via 2's complement of B getting added to A.**

# More on the 4-bit adder-subtractor

- As can be seen from the circuit, there are two outputs namely C and V (on the left-hand side).
- If the two binary numbers are unsigned, then the carry output (C) detects a carry after addition or a borrow after subtraction.
- If the numbers are instead signed, then the V (output) bit detects an overflow. If V = 0, then no overflow has occurred and the n-bit result is correct. If V =1, then the result of the operation contains an extra bit (total of n + 1 bits ) but only the rightmost n bits of the number fit into the available space indicating that an overflow has occurred.

# Textbook and Reference

- **M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018**
- **C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021**